

# An Analysis of Factors Influencing Student Academic Performance

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**Abstract**—This project investigates the factors that influence student academic performance and evaluates whether machine learning models can reliably predict final outcomes using real-world educational data. Two publicly available datasets were analyzed: one from the UCI Machine Learning Repository and one from Mendeley Data. Statistical analyses using two-sample Z-tests showed that gender, extracurricular participation, and study time were significantly associated with performance, while income, absences, and academic support variables did not demonstrate meaningful differences between groups. Three supervised learning models—Logistic Regression, Decision Tree, and Random Forest—were trained to predict student pass/fail outcomes, with the Random Forest model achieving the highest accuracy. The results demonstrate that early-term grades serve as strong predictors of final academic performance and highlight the value of data-driven approaches for identifying at-risk students and informing targeted educational interventions.

**Keywords**—Student performance, machine learning, classification models, Z-test analysis, academic achievement, Random Forest, Logistic Regression, Decision Tree, predictive modeling, educational analytics

## I. INTRODUCTION

Understanding the factors that influence student academic performance is an important topic in education, as achievement outcomes affect long-term opportunities and overall academic success. With the growing availability of open educational datasets, it has become easier to explore patterns in student learning using real-world data. Prior work in educational data mining suggests that academic outcomes are shaped by multiple variables, including demographic characteristics, study habits, parental involvement, and school-related conditions [1]. Analyses conducted on widely used datasets, such as the UCI Student Performance dataset, consistently show that factors like prior grades, attendance, and family background are closely associated with student achievement [2]. These findings highlight the value of data-driven approaches for understanding how student behavior and environment relate to academic results.

This project applies those ideas by using two credible existing datasets to examine how personal, social, and school-related characteristics connect to student performance. Rather than conducting original data collection, the work focuses on exploring real datasets through descriptive analysis and predictive modeling to identify meaningful patterns. Machine learning techniques, including supervised regression and classification models, are used to evaluate whether student performance can be predicted based on variables such as prior exam scores and study habits. By integrating these analytical approaches, the project aims to provide a clearer picture of the factors that contribute to academic outcomes and illustrate how existing educational data can be used to support informed decision-making.

## II. DATASETS

### A. Source of dataset

This project uses two publicly available and credible datasets commonly referenced in educational data mining research. The first dataset, *Student Performance* (“student-mat.csv”), originates from the UCI Machine Learning Repository [2], a well-established, peer-reviewed source widely used in academic research and coursework. The dataset was originally created by Cortez in 2008 as part of a study designed to analyze secondary school performance in Portugal. Data were collected directly from school administrative records and student surveys, ensuring that the dataset represents authentic educational information. Its long-standing presence in the UCI repository and frequent use in published studies reinforce its reliability and research value.

The second dataset, *Student Performance Metrics Dataset* (“ResearchInformation3.csv”), was obtained from Mendeley Data [1], a reputable open-access data platform that hosts peer-reviewed and researcher-submitted datasets. This dataset was created as part of a structured academic study focused on gathering information about students’ demographic characteristics, academic behaviors, and performance indicators. It includes survey-based responses collected systematically by the original researchers, making it suitable for descriptive and correlation-based analyses. Both datasets were selected because they provide rich, multi-variable information relevant to understanding student academic performance and are sourced from platforms recognized for data transparency and credibility.

### B. Character of the datasets

Both datasets in this project are provided in CSV format, which makes them easy to load and analyze using Python. The first dataset, *student-mat.csv* from the UCI Machine Learning Repository [2], contains 649 rows and 33 columns. These columns include a mixture of demographic information, family background, study habits, school-related details, and three important academic performance variables: G1, G2, and G3. These grade columns represent the first-period, second-period, and final grades, each recorded on a 0–20 point scale. Some features are numerical, such as absences, while others are categorical, such as parental education, school type, or whether the student has internet access. Because it includes numeric grade outcomes and many predictive features, the UCI dataset is the one used for the machine learning portion of this project.

The second dataset, *ResearchInformation3.csv* from Mendeley Data, is also organized in rows and columns but mainly contains survey-based information about student demographics, behaviors, and academic habits. Many of its attributes are categorical (such as yes/no responses) or self-reported measures. While it is useful for descriptive analysis and understanding general trends, it does not provide the same structured numeric grade outcomes as the UCI dataset.

For this reason, it is used only for descriptive and exploratory analysis, not for building predictive models.

Before analysis, I cleaned each dataset by checking for missing values, correcting inconsistent text labels, and converting categorical variables into numerical codes when necessary. No unit conversions were required because all numerical values were already in understandable formats. I also created a new binary category in the UCI dataset—such as classifying students as pass or fail based on their final grade—to support classification models. Since the two datasets have different structures and different types of variables, I analyzed them separately rather than combining them.

These preparation steps helped ensure both datasets were properly organized for descriptive work, and the UCI dataset was ready for use in machine learning.

### III. METHODOLOGY

In this project, the main goal of the methodology is to build a classification model that predicts whether a student will pass or fail based on the features in the UCI Student Performance dataset. To create a classification problem, the final grade (G3) was converted into a binary label, where higher scores represent a passing outcome and lower scores represent failing. Before applying machine learning, the dataset was cleaned by encoding categorical variables, checking for missing values, and selecting important features. The data was then split into training and testing sets to fairly evaluate the performance of the models. All classification methods were implemented using the scikit-learn Python library, which provides simple, reliable tools for training and testing models. The following subsections describe the classification models used in this project.

#### A. Logistic Regression

Logistic Regression is a commonly used classification method that predicts the probability that a given input belongs to a particular class. The model assumes a linear relationship between the input features and the log-odds of the outcome. The core logistic function is:

$$P(\text{pass}) = \frac{1}{(1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)})} \quad (1)$$

The function creates an output between 0 and 1, which is interpreted as the probability that the student passes.

Advantages:

- Simple and easy to interpret
- Fast to train
- Works well as a baseline

Disadvantages:

- Assumes linear relationships
- May not capture complex patterns
- Can be affected by highly correlated features

Logistic Regression was implemented using the LogisticRegression class in scikit-learn. Numerical features were scaled when needed to help with model performance.

#### B. Random Forest Classifier

Random Forest is an ensemble method that builds many decision trees and combines their predictions. Each tree is trained using random subsets of the data and features, which helps the model capture complex relationships while reducing overfitting.

Advantages:

- Handles both categorical and numerical features well
- Good at modeling complex, non-linear relationships
- Provides feature importance insights
- More robust than a single decision tree

Disadvantages:

- Less interpretable
- More computationally expensive
- Requires tuning for best performance

Random Forest was chosen because student success often depends on many interacting factors, and this model can handle that complexity. It was implemented using RandomForestClassifier from scikit-learn.

#### C. Decision Tree Classifier

A Decision Tree is a simple classification model that splits data into branches based on feature values. Each split tries to separate the data into more pure groups, leading to a final prediction at the leaf nodes. Decision Trees do not assume any linear relationship and can capture more flexible patterns than Logistic Regression.

Advantages:

- Very easy to visualize and interpret
- Handles categorical and numerical data
- Captures non-linear relationships
- No need for feature scaling

Disadvantages:

- Can easily overfit the training data
- Sensitive to small changes in the dataset
- Often less accurate than ensemble methods like Random Forest

Decision Trees were included because they help show the structure of the data and make it easier to understand which features are most important. They also serve as the base model for ensemble methods like Random Forest. I implemented the model using DecisionTreeClassifier from scikit-learn.

### IV. RESULTS

This section presents the findings related to each research question using two-sample Z-tests. The results evaluate how demographic, socioeconomic, and school-related factors are associated with student performance and assess whether performance outcomes can be predicted using machine learning methods. All Z-tests were conducted at the 0.05 significance level.

### A. Factors Associated With Student Performance

A Z-test comparing overall performance between male and female students revealed a statistically significant difference,  $z = -3.596$ ,  $p = 0.0003$ . Female students achieved significantly higher performance scores than male students, suggesting an association between gender and academic outcomes in this dataset.

A second Z-test examined whether participation in extracurricular activities was associated with performance. The results showed a strong significant difference,  $z = 7.320$ ,  $p < 0.0001$ , indicating that students involved in extracurricular activities scored higher on average than those who were not. This suggests that extracurricular engagement may play a positive role in student learning and academic success.

Income level (converted to a binary low vs. higher-income variable) was also analyzed using a Z-test. The results were not statistically significant,  $z = 0.703$ ,  $p = 0.4822$ , indicating no evidence of a meaningful difference in academic performance between lower-income students and their higher-income peers. This suggests that, within this dataset, income differences did not strongly influence overall performance once reduced to two broad categories.

### B. School-Related Factors and Academic Outcomes

School-related behaviors were examined using Z-tests on binarized versions of study time, absences, and support variables. Students with higher study time showed significantly higher final grades (G3) than those with lower study time,  $z = -2.267$ ,  $p = 0.0234$ . This result aligns with expectations that increased study time positively contributes to academic performance.

A Z-test comparing students with low versus high levels of absences did not show a significant difference in final grades,  $z = 1.524$ ,  $p = 0.1276$ . This suggests that absence levels, when grouped into two broad categories, did not strongly distinguish student performance in this dataset.

School support also did not show a statistically significant difference between groups,  $z = -1.647$ ,  $p = 0.0996$ , though the result approached significance. Students receiving school support tended to score slightly lower, which may reflect that academic support programs are typically offered to students already experiencing academic challenges.

Similarly, family support was not significantly associated with performance,  $z = 0.777$ ,  $p = 0.4372$ , indicating that students reporting family academic support did not differ meaningfully from those without such support.

### C. Machine Learning Classification Results

To assess whether student performance could be predicted using study habits and prior academic results, three supervised learning models were trained on the UCI student-mat dataset: Logistic Regression, Decision Tree, and Random Forest Classifier. All models used the same set of features (studytime, failures, absences, G1, G2) to predict a binary pass/fail outcome based on the final grade (G3).

- The Logistic Regression model achieved an accuracy of 0.8987, with strong performance for both pass and fail classes (precision = 0.96 for pass, 0.81 for fail).

- The Decision Tree model achieved a similar overall accuracy of 0.8987, performing slightly better on recall for the pass class (0.92).
- The Random Forest Classifier performed the best overall, with an accuracy of 0.9114 and strong balanced performance across all metrics (precision = 0.96 for pass, 0.83 for fail).

## V. DISCUSSION

Across all models, the results show that early-term grades (G1 and G2) were strong predictors of final performance. This finding reflects a common pattern in educational settings where past academic performance is a reliable indicator of future outcomes. However, relying heavily on these features introduces some limitations. Because G1 and G2 directly contribute to the final grade (G3), models may achieve high accuracy without fully capturing other important influences on student performance such as motivation, study habits, or support systems. As a result, the models may not generalize well to situations where early-term grades are unavailable, incomplete, or themselves influenced by external factors.

Another limitation involves the way several variables were transformed for statistical testing and modeling. Converting continuous or multi-level categorical variables—such as income, absences, and study time—into binary groups simplifies the analysis but may obscure more subtle relationships. This simplification may explain why certain factors, such as absences or support services, did not show statistically significant effects despite being identified as important in prior educational literature. Additionally, the presence of potential selection bias in school support programs—where assistance is often provided to struggling students—may have influenced the observed differences.

For the decision tree model, model instability and potential overfitting represent another limitation. Single decision trees tend to memorize patterns in the training data, which may reduce their ability to generalize to new data. In contrast, the Random Forest model performed better by averaging the results of multiple trees, thereby reducing variance and improving stability. Still, further improvements could be achieved by tuning hyperparameters, experimenting with additional models, or evaluating performance using cross-validation.

Future work could explore more detailed encodings of socioeconomic and behavioral variables, including multi-level groupings or continuous scales. Incorporating additional predictors, such as learning engagement or teacher feedback, may improve both interpretability and model performance. Expanding the analysis to different datasets or predicting continuous final grades instead of binary outcomes could provide deeper insights into the mechanisms that influence student achievement.

## VI. CONCLUSION

This project examined the factors that influence student academic performance and evaluated whether performance can be predicted using machine learning models. The statistical analysis showed that gender, extracurricular participation, and study time had significant effects on academic outcomes, while income, absences, and support services did not show meaningful differences when simplified into two groups. The machine learning models achieved high accuracy, with the Random Forest classifier performing the

best, demonstrating that early-term grades are strong predictors of final performance. These results highlight the importance of early academic monitoring and student engagement in improving achievement. In real-world educational settings, such insights can support the development of targeted interventions that identify at-risk students early and promote practices—such as consistent study habits and involvement in learning activities—that enhance long-term academic success.

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#### REFERENCES

- [1] T. Hasan, M. M. Hasan, and T. Manzoor, "Student performance metrics dataset," Mendeley Data, V1, 2024, doi: 10.17632/5b82ytz489.1.
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